

ECONOMETRICS I

Problem Set 3

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1. **Argue extensively** whether each of the following statements is true, false, or uncertain based on technical arguments, graphs, and/or formulas.

- (a) Regressions with excess regressors tend to be inefficient. However, the exclusion of certain variables could also lead to high variance.

Solution: True. For the coefficient $\hat{\beta}_j$ we have $\widehat{\text{var}}[\hat{\beta}_j|x] = \hat{\sigma}^2 \frac{1}{n} \frac{1}{1-R_j^2} \frac{1}{\widehat{\text{var}}[x_j]}$. The inclusion of very important variables can reduce $\hat{\sigma}^2 = \hat{u}'\hat{u}/(n-k)$. But if these are irrelevant, it increases both $\hat{\sigma}^2$ and $VIF = \frac{1}{1-R_j^2}$, especially when there is high multicollinearity, which is reflected in a high R_j^2 .

- (b) In a univariate OLS regression satisfying all necessary conditions, the more the regressor varies in repeating sampling, the better (more efficient) the estimation will be.

Solution: False or uncertain. It is true that the variance formula of the slope in a simple regression is

$$\text{var}(\hat{\beta}_2|x) = \frac{\sigma^2}{\sum (x_i - \bar{x})^2} = \frac{\sigma^2}{n \text{var}(x)},$$

where a higher variance $\text{var}(x)$ of the regressor means a lower variance of the slope estimate in repeated sample. However, $\text{var}(x)$ is the dispersion of x around its mean while holding the vector x *fixed* from sample to sample. The notation “ $|x$ ” means that x is *given* (i.e. fixed) from sample to sample. We did not study what happens if x varies randomly instead.

- (c) If we obtain a coefficient not significantly different from zero in a regression, we

should conclude that this regressor is not relevant in the data generating process.

Solution: False. Even if the assumptions of the classical model were satisfied, a relevant variable could be non-significant when there are few observations, when the error variance is very high, or when there is high multicollinearity (a high VIF). Moreover, non-significance may be an artifact of our violation of some the assumptions of the classical model (for example, omitted variable biases or non-normal error variance).

- (d) The p-value of a coefficient is a better measure of precision than the standard deviation of a coefficient.

Solution: Ambiguous and rather false. If the assumptions of the classical model are satisfied, the standard deviation of a coefficient is directly a measure of precision, while the p-value commonly corresponds to the probability that the estimated coefficient is as far or farther from zero when its population counterpart is zero. The latter is not necessarily a measure of precision, because a precisely estimated coefficient could be non-significant.

- (e) Homoskedasticity in the data generating process is an essential assumption ensuring the the unbiasedness of the OLS estimator.

Solution: False. What is needed to hold in the data generating process is $y = X\beta + u$ and $\mathbb{E}[u|X] = 0$:

$$\begin{aligned}\hat{\beta} &= (X'X)^{-1}X'y \\ &= (X'X)^{-1}X'(X\beta + u) && \text{given } \boxed{y = X\beta + u} \\ &= \beta + (X'X)^{-1}X'u \\ \mathbb{E}(\hat{\beta}|X) &= \beta + \mathbb{E}[(X'X)^{-1}X'u|X] \\ &= \beta + (X'X)^{-1}X'\mathbb{E}[u|X] \\ &= \beta && \text{given } \boxed{\mathbb{E}[u|X] = 0}\end{aligned}$$

- (f) When estimating a regression that includes only irrelevant or non-causal variables, it is impossible to obtain significant coefficients.

Solution: False. It could occur that an irrelevant regressor correlates with an omitted relevant regressor, acting as a proxy for it. Spurious correlation could generate a statistically significant relationship.

- (g) When assuming $\mathbb{E}[u|X] = 0$, one implicitly assumes that all variables excluded from a regression (relevant or irrelevant) do not correlate with the variables included in it.

Solution: The statement is correct except for the mention of irrelevant variables. If an irrelevant variable is excluded, this does not represent a violation of the assumption $\mathbb{E}[u|X] = 0$. In contrast, if a relevant variable were omitted, then it would be implicitly included in u and with $\mathbb{E}[u|X] = 0$ it is required that there be no correlation between that omitted variable and the regressors included in the regression.

- (h) Vertical and horizontal outliers are influential in the slopes of an OLS estimation in different ways because the estimator minimizes vertical residuals and not horizontal ones.

Solution: True. Influence requires leverage, which does not occur in vertical outliers (y-axis) but rather in horizontal outliers (X-axes). The OLS estimator minimizes the sum of squared residuals (u_i), defining these as the vertical distance between the value of the dependent variable (y_i) for each observation and the forecast value of the regression function ($\hat{y}_i$).

2. For each concept in the left column, indicate all related concepts from the right column.

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|---|--|
| (i) Low number of observations | (A) Unreliable coefficients |
| (ii) High multicollinearity | (B) Inefficiency |
| (iii) Omission of relevant variable not related to regressors | (C) High variance of the error term |
| (iv) Omission of relevant variable related to regressors | (D) Bias |
| (v) Inclusion of irrelevant variable | (E) Invalidity of the coefficient variance-covariance matrix |
| (vi) Heteroscedasticity | (F) Invalidity of a projection variance-covariance matrix |
| (vii) 'Horizontal' outliers | (G) Invalid confidence interval |
| (viii) Simultaneous causality | (H) Coefficient vector not defined |
| (ix) Non-normality of errors | |

Solution:

- (i) Low number of observations $\rightarrow$ B and, therefore, A. C does not apply if we refer to the DGP, but it does when we estimate $\hat{\sigma}^2$, due to the denominator $n - k$.
- (ii) High multicollinearity $\rightarrow$ B and, therefore, A.
- (iii) Omission of relevant variable not related to regressors $\rightarrow$ If the variable is very relevant: C and, therefore, B and A.
- (iv) Omission of relevant variable related to regressors $\rightarrow$ Directly D and C, implying A, E, F, G.
- (v) Inclusion of irrelevant variable $\rightarrow$ If it generates high multicollinearity: B and, therefore, A.
- (vi) Heteroscedasticity $\rightarrow$ E, F, G.
- (vii) High influence $\rightarrow$ A. Due to erroneous estimation of the error: B, C, E, F, G. Note that bias does not apply as it is a concept defined in the context of repeated sampling.
- (viii) Simultaneous causality $\rightarrow$ Mainly D. From this derives A, E, F, G.
- (ix) Non-normality of errors $\rightarrow$ G.